Counting to Four is still a Chore for VLMs

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Research Questions

- Do VLMs count based on what they are truly seeing or on their learned priors, and in which component does this occur?
- Where exactly does visual evidence degrade in the VLM architecture during counting tasks?
- Can attention redistribution interventions improve visual grounding during counting?

Core Idea

VLMs fail at basic counting despite excelling at complex reasoning because they suffer from a "visual attention sink" - during generation, they allocate ~90% attention to text tokens and only ~10% to visual tokens, relying on linguistic priors rather than visual evidence. The key insight is that counting-relevant visual information is actually present in early vision layers but gets progressively diluted as it passes through the language model layers, where textual biases dominate. The authors' MAS (Modality Attention Share) intervention forces models to maintain a minimum visual attention budget during answer generation, acting as a regularization constraint that prevents complete drift toward text-based reasoning.

Key Formula

Modality Attention Share

$$MAS_\ell = \frac{1}{|T|} \sum_{t \in T} \frac{\sum_h \sum_{j \in V} A_{t \to j}^{(h,\ell)}}{\sum_h \sum_{j \in V \cup X} A_{t \to j}^{(h,\ell)}}$$

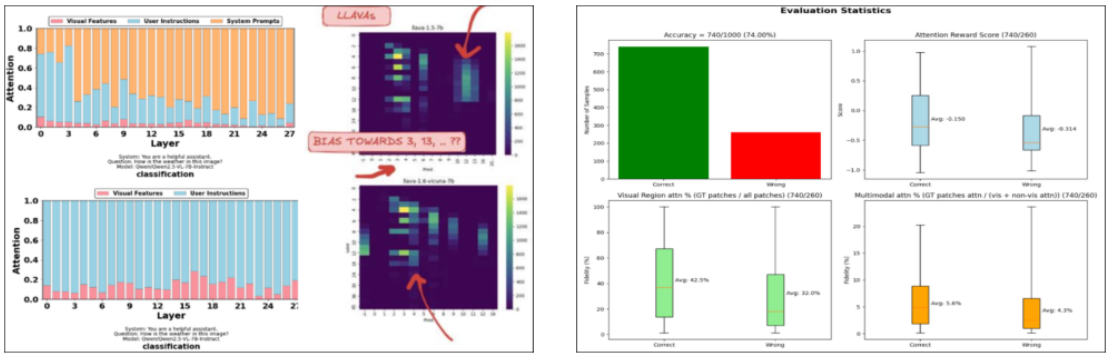
Computes the ratio of attention mass allocated to visual tokens V versus all tokens (visual V + textual X) at layer ℓ, averaged over decoding steps T. Used to quantify visual attention budget during generation.

Visual Constraint Loss

$$L_{mas} = \max(0, \tau - MAS)$$

Hinge loss that activates when visual attention MAS drops below safety threshold τ. Forces models to maintain minimum visual consultation during answer generation, preventing complete drift to textual priors.

Key Figure



(a) **Modality Attention Share.** Despite the task being purely visual, attention mass is overwhelmingly allocated to text tokens (blue bars), leaving only $\sim 10.7\%$ for visual tokens (red bars).

(b) **Grounding Statistics.** Even for correct predictions (green), the Visual Region Attention % is low ($\sim 42.5\%$) and the Attention Reward Score is negative (-0.15), indicating weak pixel commitment.

Figure 3. **Evidence of Text-Prior Dominance.** Our probing analysis reveals that models suffer from a "visual attention sink," where computation drifts away from image tokens toward system prompts and instructions.

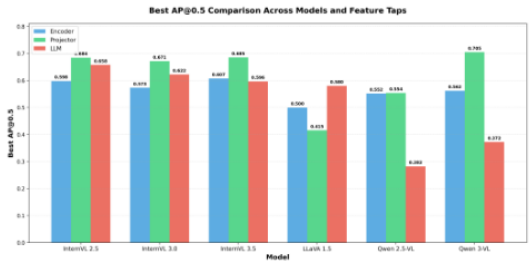


Figure 4. **Spatial evidence is strong in the vision stack but fades in the language layers.** Detection performance (AP@0.5) peaks at the projector tap (green) and drops at the LLM tap (red) across modern architectures, quantifying the loss of spatial information.

performance difference reflects the quality of the retained representation, not the probe's learning capacity.

features at this stage are not only weaker but also structurally noisier.

Quantifying signal fade from vision to language. Despite an identical probing setup, we observe a consistent drop in detection performance from the projector to the LLM across multiple architectures (Figure 4). Average Precision (AP) peaks at the projector tap but degrades at the LLM tap. For example, in Qwen2.5-VL, AP falls from $\mathbf{0.554}$ (projector) to $\mathbf{0.282}$ (LLM); in Qwen3-VL, from $\mathbf{0.705}$ to $\mathbf{0.372}$. These results indicate that while the visual backbone encodes object-level spatial information well, this signal is diluted during fusion and language reasoning.

Observation 4. *Spatial evidence is strong in the vision stack but fades in the language layers. The visual backbone encodes objects well (high AP at the projector), yet this spatial identity is largely lost during LLM reasoning (low AP at the LLM), indicating a failure of integration rather than perception.*

Figure 4: Shows quantitative evidence that spatial detection performance (AP@0.5) peaks at the projector stage but consistently drops at LLM layers across multiple architectures, proving visual information degradation.

Limitations

- MAS improvements are modest (up to 1% accuracy gain) and backbone-dependent, with mixed effects across different architectures
- Single fixed threshold τ=0.4 may not be optimal for all tasks or models, requiring task-specific tuning
- Intervention only addresses attention allocation, not the fundamental issue of spatial structure preservation in language layers
- Results show trade-offs between in-distribution validation gains and out-of-distribution generalization performance